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PIXES: Protocol Implementing Xarxes Encapsulation and Sockets

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Abstract

PIXES is an experimental Layer 2 tunnelling protocol that encapsulates Ethernet frames inside UDP datagrams, enabling geographically separated hosts to share a common virtual Ethernet segment. A PIXES deployment consists of one server, acting as a virtual Ethernet switch, and one or more clients, each of which is attached to a kernel TAP device. The server learns client MAC addresses dynamically and forwards frames accordingly. This document specifies the PIXES wire format, the connection lifecycle, the frame-forwarding rules, and the session- management mechanisms.

Resum

Resum en català

PIXES és un protocol experimental de tunelització de capa 2 que encapsula trames Ethernet dins de datagrames UDP, permetent que màquines separades geogràficament comparteixin un segment Ethernet virtual comú. Un desplegament PIXES consta d'un servidor, que actua com un commutador Ethernet virtual, i un o més clients, cadascun connectat a un dispositiu TAP del nucli Linux. El servidor aprèn les adreces MAC dels clients de manera dinàmica i reenvía les trames corresponentment. Aquest document especifica el format de cabdal del protocl PIXES, el cicle de vida de la connexió, les regles de reenviament de trames i els mecanismes de gestió de sessions.

Taula de Continguts / Table of Contents

1. Introduction 3
2. Conventions 3
3. Terminology 4
4. Packet Format 5
   4.1. PIXES Header 5
   4.2. PIXES Traffic Packet 6
   4.3. Payload Semantics by Opcode 7
5. Message Types 8

5.1. REGISTER 8
5.2. AUTH 9
5.3. TRAFFIC 9
5.4. KEEPALIVE 10
5.5. ACK 10
5.6. REJECT 11
6. Client State Machine 11
7. Server Session State Machine 14
8. Frame Forwarding 16
8.1. MAC Address Learning 16
8.2. Forwarding Rules 17
8.3. Unknown Unicast Policy 17
9. Security Considerations 18
10. IANA Considerations 18
11. References 19
11.1. Normative References 19
11.2. Informative References 19
Appendix A. Authors' Addresses 20

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1. Introduction

Introducció

Modern network virtualisation frequently requires that hosts connected to physically distinct networks communicate as though they were on the same Ethernet segment. PIXES provides a lightweight mechanism to achieve this by tunnelling raw Ethernet frames over UDP.

A PIXES deployment has two roles:

- **Client:** A host that opens a kernel TAP device, reads Ethernet frames produced by the local network stack, encapsulates them in PIXES packets, and transmits them to the server via UDP. When the server forwards a PIXES packet to the client, the client extracts the embedded Ethernet frame and injects it back into the TAP device.
- **Server:** A host that receives PIXES packets from multiple clients, learns the Ethernet MAC addresses of those clients, and forwards frames to the correct destination client using standard Ethernet switching logic.

PIXES operates entirely at Layer 2. IP addressing and routing above the Ethernet layer are outside the scope of this protocol.

The key design goals of PIXES are simplicity, ease of implementation in low-level languages such as C, and pedagogical clarity. PIXES intentionally omits features such as encryption, compression, and dynamic port negotiation.

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2. Conventions

Convencions

The key words “MUST”, “MUST NOT”, “REQUIRED”, “SHALL”, “SHALL NOT”, “SHOULD”, “SHOULD NOT”, “RECOMMENDED”, “MAY”, and “OPTIONAL” in this document are to be interpreted as described in RFC 2119 [RFC2119].

All multi-octet numeric fields in PIXES packets are transmitted in network byte order (big-endian), with the most significant octet first, unless explicitly stated otherwise.

---

3. Terminology

Terminologia

TAP Device: A kernel virtual network interface that operates at Layer 2 (Ethernet). A user-space process reads and writes raw Ethernet frames through a TAP device file descriptor. In Linux, TAP devices are accessed via `/dev/net/tun` with the `IFF_TAP` flag.

VPN Client (or Client): A host running the PIXES client software. The client binds a TAP device to a PIXES session and forwards frames between the TAP device and the server via UDP.

VPN Server (or Server): A host running the PIXES server software. The server acts as a virtual Ethernet switch: it receives PIXES packets from clients, learns MAC addresses, and forwards frames to the appropriate destination client.

Session: The logical association between the server and a specific client, identified by the client’s Client Identifier. A session is created upon receipt of a REGISTER message and destroyed upon timeout or explicit disconnection.

Client Identifier (CID): An unsigned 16-bit integer (0–65535) that uniquely identifies a client within a PIXES server. The CID is chosen by the client operator and MUST be unique among all concurrently connected clients.

Ethernet Frame: A Layer 2 Protocol Data Unit as defined in IEEE 802.3. In PIXES, an Ethernet frame consists of a 6-octet destination MAC address, a 6-octet source MAC address, a 2-octet EtherType field, and a variable-length payload. The minimum Ethernet frame carried by PIXES is 14 octets.

MAC Address: A 48-bit (6-octet) hardware address identifying a network interface at Layer 2. MAC addresses are written in this document as groups of two hexadecimal digits separated by hyphens, e.g., `AA-BB-CC-DD-EE-FF`.

Keepalive: A periodic control message sent by the client to signal that the session is still active, preventing the server from expiring the session due to inactivity.

Session Timeout: A server-configured duration (in seconds) after which a session with no received packets is considered expired and MUST be removed.

Sequence Number: An unsigned 64-bit monotonically increasing counter maintained per client, incremented for each TRAFFIC packet sent.

---

4. Packet Format

Format de Paquet

4.1. PIXES Header

Capçalera PIXES

Every PIXES packet begins with a fixed-size header of exactly 11 octets. The header is structured as follows:

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|    Opcode     |         Client Identifier     |  Payload[0]   |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  Payload[1]   |  Payload[2]   |  Payload[3]   |  Payload[4]   |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  Payload[5]   |  Payload[6]   |  Payload[7]   |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

Opcode (8 bits): Identifies the type of PIXES message. Valid values are defined in Section 5. A receiver that receives an unknown Opcode value MUST silently discard the packet.

Client Identifier (16 bits, network byte order): The CID of the client that originates or is the intended target of the packet. In client-to-server messages, this field contains the sending client's CID. In server-to-client messages, this field contains the destination client's CID.

Payload (64 bits / 8 octets): A fixed-size field whose meaning depends on the Opcode. For opcodes that do not use the payload, all 8 octets MUST be set to zero by the sender and MUST be ignored by the receiver.

Any PIXES packet shorter than 11 octets MUST be silently discarded by the receiver.

4.2. PIXES Traffic Packet

Paquet de Tràfic PIXES

A PIXES TRAFFIC packet carries a single encapsulated Ethernet frame. It consists of the standard 11-octet PIXES header followed immediately by the raw

Ethernet frame octets. There is no length field; the encapsulated frame length is derived from the UDP datagram length minus 11 octets.

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  Opcode=0x03  |         Client Identifier     |  SeqNum[0]    |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  SeqNum[1]    |  SeqNum[2]    |  SeqNum[3]    |  SeqNum[4]    |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  SeqNum[5]    |  SeqNum[6]    |  SeqNum[7]    |               |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+               +
|             Ethernet Frame (variable length)                  |
|                              ...                              |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

SeqNum[0..7] (64 bits, network byte order): A monotonically increasing sequence number, starting at zero and incremented by one for each TRAFFIC packet sent. The sequence number occupies the Payload field of the PIXES header. The receiver SHOULD accept packets regardless of sequence number order; PIXES does not guarantee in-order delivery.

Ethernet Frame: The raw Ethernet frame beginning at octet offset 11 of the UDP payload. The frame MUST be at least 14 octets in length (6-octet destination MAC + 6-octet source MAC + 2-octet EtherType). A TRAFFIC packet whose embedded frame is shorter than 14 octets MUST be silently discarded by the server.

4.3. Payload Semantics by Opcode

Semàntica del Camp Payload per Opcode

The following table summarises the meaning of the 8-octet Payload field for each Opcode value.

Opcode	Name	Payload Contents
0x01	REGISTER	8 zero octets
0x02	AUTH	8-octet password, ASCII [A-Za-z0-9], no NUL
0x03	TRAFFIC	64-bit sequence number, network byte order
0x04	KEEPALIVE	8 zero octets
0x05	ACK	8 zero octets
0x06	REJECT	8 zero octets

5. Message Types

Tipus de Missatge

The following table provides a summary of all PIXES message types.

Value	Name	Direction	Payload
0x01	REGISTER	Client -> Server	8 zero octets
0x02	AUTH	Client -> Server	8-octet ASCII password
0x03	TRAFFIC	Bidirectional	64-bit sequence number
0x04	KEEPALIVE	Client -> Server	8 zero octets
0x05	ACK	Server -> Client	8 zero octets
0x06	REJECT	Server -> Client	8 zero octets

5.1. REGISTER (0x01)

Registre

The REGISTER message is sent by the client to initiate a new session with the server.

Sender behaviour: The client MUST send a REGISTER message as the first step of the connection procedure. The Payload field MUST be set to 8 zero octets. The client MUST then wait up to 5 seconds for an ACK or REJECT response. If no valid response is received within 5 seconds, the client MUST treat the registration as failed and SHOULD terminate with an error.

Receiver behaviour: Upon receipt of a REGISTER message, the server MUST create a new session entry for the specified CID (or replace an existing one) and set the session state to REGISTERING. If the CID is already associated with a different source address, the server SHOULD log the address change and remove all MAC table entries associated with the previous session. The server MUST respond with an ACK message.

5.2. AUTH (0x02)

Autenticació

The AUTH message is sent by the client to authenticate an established session using a shared password.

Sender behaviour: The client MUST send an AUTH message after receiving an ACK in response to REGISTER. The Payload field MUST contain exactly 8 octets of ASCII characters in the set `[A-Za-z0-9]`. The client MUST wait up to 5 seconds for an ACK or REJECT response per attempt. If a REJECT is received or the timeout elapses, the client MAY retransmit the AUTH message.

The client MUST NOT retransmit AUTH more than 3 times in total. If all 3 attempts fail, the client MUST terminate with an error.

Receiver behaviour: Upon receipt of an AUTH message, the server MUST look up the session for the specified CID. If no session exists, or the session is not in the REGISTERING state, the server MUST respond with REJECT. If the provided password matches the configured credential for the session, the server MUST transition the session to the AUTHENTICATED state and respond with ACK. Otherwise, the server MUST respond with REJECT and the session MUST remain in the REGISTERING state.

5.3. TRAFFIC (0x03)

Tràfic

The TRAFFIC message carries a single encapsulated Ethernet frame between client and server, or from server to client.

Sender behaviour (client): A client in the ESTABLISHED state MAY send a TRAFFIC message whenever a frame is available on its TAP device. The Payload field MUST contain a 64-bit sequence number encoded in network byte order, starting at zero and incremented by one per packet. The Ethernet frame MUST be appended immediately after octet 10 of the PIXES header.

Sender behaviour (server): When forwarding a frame to a client, the server MUST transmit the complete original UDP payload (PIXES header plus Ethernet frame) to the destination client unchanged. The server MUST NOT modify the Opcode, Client Identifier, Sequence Number, or frame contents.

Receiver behaviour (client): Upon receipt of a TRAFFIC message, the client MUST extract the Ethernet frame (octets 11 onward of the UDP payload) and write it to the local TAP device. The client MUST silently discard TRAFFIC packets shorter than 12 octets (header plus at least one frame octet).

Receiver behaviour (server): The server MUST silently discard TRAFFIC messages received from clients not in the AUTHENTICATED state. For authenticated clients, the server MUST process the frame as described in Section 8. The server MUST update the `last_seen` timestamp of the session upon receipt of any TRAFFIC message.

5.4. KEEPALIVE (0x04)

Keepalive

The KEEPALIVE message is sent by the client to signal that the session remains active in the absence of data traffic.

Sender behaviour: A client in the ESTABLISHED state MUST send a KEEPALIVE message if no TRAFFIC message has been sent within the preceding 10 seconds. The Payload field MUST be set to 8 zero octets.

Receiver behaviour: Upon receipt of a KEEPALIVE from a client in the AUTHENTICATED state, the server MUST update the `last_seen` timestamp of the session. The server MUST NOT send any response to a KEEPALIVE. A KEEPALIVE received from a client not in the AUTHENTICATED state MUST be silently discarded.

5.5. ACK (0x05)

Confirmació

The ACK message is sent by the server to indicate successful processing of a REGISTER or AUTH message.

Sender behaviour: The server MUST send an ACK in response to a valid REGISTER message (see Section 5.1) and in response to a successful AUTH message (see Section 5.2). The Payload field MUST be set to 8 zero octets. The CID in the ACK MUST match the CID in the request.

Receiver behaviour: Upon receipt of an ACK in the REGISTERING client state, the client MUST transition to the AUTHENTICATING state and send an AUTH message. Upon receipt of an ACK in the AUTHENTICATING state, the client MUST transition to the ESTABLISHED state. An ACK received in any other state MUST be silently discarded.

5.6. REJECT (0x06)

Rebuig

The REJECT message is sent by the server to indicate that a REGISTER or AUTH request has been denied.

Sender behaviour: The server MUST send a REJECT in response to a REGISTER message that cannot be processed, and in response to a failed AUTH attempt (see Section 5.2). The Payload field MUST be set to 8 zero octets.

Receiver behaviour: Upon receipt of a REJECT during the registration or authentication phase, the client MUST treat the current attempt as failed and apply the retry logic defined in Sections 5.1 and 5.2 respectively. A REJECT received in the ESTABLISHED state MUST be silently discarded.

---

6. Client State Machine

Màquina d'Estats del Client

A PIXES client maintains a single connection state. The states and transitions are defined below.

6.1. States

Estats

```
+-------------+-------------------------------------------------------------+
| State       | Description                                                 |
+-------------+-------------------------------------------------------------+
| CLOSED      | No active session.  Initial state and terminal error        |
|             | state.                                                      |
| REGISTERING | A REGISTER message has been sent; the client is waiting     |
|             | for ACK or REJECT from the server.                          |
| AUTHENTICATING | An AUTH message has been sent; the client is waiting     |
|             | for ACK or REJECT.  At most 3 AUTH attempts are made.       |
| ESTABLISHED | The session is fully authenticated.  The client             |
|             | forwards frames between the TAP device and the server.      |
+-------------+-------------------------------------------------------------+
```

6.2. State Diagram

Diagrama d'Estats

```
              start
                |
                v
          +-----------+
          |  CLOSED   |<------------------------------+
          +-----------+                               |
                |                                     |
  send REGISTER |                                     |
                v                                     |
      +--------------+  recv REJECT or 5 s timeout  |
      | REGISTERING  |----------------------------->(error exit)
      +--------------+
                |
     recv ACK   |
   send AUTH    |
   (attempt=1)  |
                v
   +------------------+
   |  AUTHENTICATING  |
   +------------------+
          |        |
          |        | recv REJECT or 5 s timeout,
          |        | attempt < 3:
          |        | send AUTH (attempt++)
          |        +--------(self-loop)
          |
```

```
        | recv REJECT or 5 s timeout,
        | attempt == 3:
        +----------------------------->(error exit)
        |
        | recv ACK
        |
        v
+--------------+   recv SIGINT / SIGTERM
| ESTABLISHED  |--------------------------------> close fds, exit 0
+--------------+
        |    ^
        |    |  [events processed in select() loop]
        |    |
        +----+  TAP readable:
                   send TRAFFIC (seq++)
                UDP recv TRAFFIC (len > 11):
                   write frame to TAP
                10 s timeout:
                   send KEEPALIVE
```

6.3. Transition Descriptions

Descripció de Transicions

CLOSED → REGISTERING: Triggered when the client starts up. The client MUST open the TAP device and the UDP socket, then send a REGISTER message. The client MUST set a receive timeout of 5 seconds.

REGISTERING → ESTABLISHED path (via AUTHENTICATING): Upon receipt of ACK in the REGISTERING state, the client MUST send AUTH with the 8-octet password in the Payload field and transition to AUTHENTICATING. Upon receipt of ACK in the AUTHENTICATING state, the client MUST clear any socket receive timeout and transition to ESTABLISHED.

REGISTERING → CLOSED (error): If the registration response is REJECT, or if the 5-second timeout elapses with no valid response, the client MUST print an error message to standard error and terminate with exit code 1.

AUTHENTICATING → AUTHENTICATING (retry): If a REJECT is received or the per-attempt 5-second timeout elapses, and the total number of attempts is less than 3, the client MUST retransmit the AUTH message and restart the timeout.

AUTHENTICATING → CLOSED (error): If all 3 authentication attempts fail, the client MUST print `"Authentication failed after 3 attempts"` to standard error and terminate with exit code 1.

ESTABLISHED → CLOSED (clean shutdown): Upon receipt of SIGINT or SIGTERM, the client MUST close the TAP file descriptor and the UDP

socket, then exit with code 0.

ESTABLISHED → ESTABLISHED (data events): The client MUST use `select()` (or equivalent) to monitor both the TAP file descriptor and the UDP socket simultaneously, with a timeout of 10 seconds:

- If the TAP device is readable, the client MUST call `tap_read()`, construct a TRAFFIC packet (header with the current sequence number followed by the frame), send it via UDP, and increment the sequence number.
- If the UDP socket is readable, the client MUST receive the datagram. If the Opcode is TRAFFIC and the total length exceeds 11 octets, the client MUST write the frame payload (octets 11 onward) to the TAP device using `tap_write()`. All other opcodes received in the ESTABLISHED state MUST be silently discarded.
- If `select()` returns due to timeout (no file descriptor became readable within 10 seconds), the client MUST send a KEEPALIVE packet.

---

7. Server Session State Machine

Màquina d'Estats de Sessió del Servidor

The server maintains one independent state machine per client session, keyed by Client Identifier.

7.1. Session States

Estats de Sessió

```
+--------------+-------------------------------------------------------+
| State        | Description                                           |
+--------------+-------------------------------------------------------+
| NONE         | No session exists for this CID.  This is the implicit  |
|              | initial state before REGISTER is received, and the     |
|              | state to which an expired session returns.             |
| REGISTERING  | A REGISTER message has been received and ACK has been  |
|              | sent.  Awaiting AUTH from the client.                  |
| AUTHENTICATED| The client has been successfully authenticated.  The   |
|              | server will forward TRAFFIC to and from this session.  |
+--------------+-------------------------------------------------------+
```

7.2. Session State Diagram

Diagrama d'Estats de Sessió

```
   +---------+
   |  NONE   |<-----------------------------------------+
```

```
  +---------+                                          |
        |                                              |
        | recv REGISTER                                |
        | create session (last_seen = now)             |
        | send ACK                                     |
        v                                              |
 +--------------+                                      |
 | REGISTERING  |                                      |
 +--------------+                                      |
        |       |                                      |
        |       | recv AUTH (invalid password          |
        |       | or session not in REGISTERING):      |
        |       | send REJECT                          |
        |       +-----------(stay REGISTERING)         |
        |                                              |
        | recv AUTH (valid password)                   |
        | session.state = AUTHENTICATED                |
        | update last_seen                             |
        | send ACK                                     |
        v                                              |
+---------------+   watchdog: now - last_seen > timeout |
| AUTHENTICATED |-------------------------------------->+
+---------------+   remove session and MAC entries     |
        |       |                                      |
        |       | recv REGISTER (re-registration):     |
        |       | replace session, send ACK            |
        |       +-----------(back to REGISTERING)------->+
        |                                              ^
        | recv TRAFFIC: process frame (Section 8)      |
        |   update last_seen                           |
        | recv KEEPALIVE: update last_seen             |
        +----------------(stay AUTHENTICATED)
```

7.3. Transition Descriptions

Descripció de Transicions

NONE → REGISTERING: Upon receipt of a REGISTER message, the server MUST create a new session entry for the CID. If a session already exists for that CID with a different source address, the server SHOULD log the change and MUST remove all MAC table entries associated with the old session. The server MUST set `last_seen` to the current monotonic time and MUST respond with ACK.

REGISTERING → AUTHENTICATED: Upon receipt of an AUTH message where the session exists, is in the REGISTERING state, and the Payload contains the correct 8-octet password, the server MUST transition the session to

AUTHENTICATED, update `last_seen`, and respond with ACK.

REGISTERING → REGISTERING (AUTH failure): If the AUTH password is incorrect, or if the session is not in the REGISTERING state, the server MUST respond with REJECT. The session state MUST NOT change.

AUTHENTICATED → NONE (watchdog expiry): The server MUST run a periodic watchdog process. Any session for which `current_time - last_seen` exceeds the configured session timeout MUST be removed. All MAC table entries associated with the expired CID MUST be removed simultaneously. The server SHOULD log the expiry event.

AUTHENTICATED → NONE (re-registration): If a REGISTER message is received for a CID that is already in the AUTHENTICATED state, the server MUST treat it as a fresh registration: remove existing MAC entries for that CID, create a new session in the REGISTERING state, and respond with ACK.

---

8. Frame Forwarding

Reenviament de Trames

8.1. MAC Address Learning

Aprenentatge d'Adreces MAC

The server maintains a MAC learning table that maps 6-octet Ethernet MAC addresses to Client Identifiers.

Upon receipt of a TRAFFIC message from a client in the AUTHENTICATED state, the server MUST extract the source MAC address from the encapsulated Ethernet frame. The source MAC address occupies octets 6–11 (inclusive, zero-indexed) of the Ethernet frame, i.e., octets 17–22 of the UDP payload. The server MUST associate this MAC address with the sending client's CID in the MAC learning table, replacing any previous association for that MAC address.

When a session is removed (either by expiry or re-registration), the server MUST remove all MAC table entries that were associated with that session's CID.

8.2. Forwarding Rules

Regles de Reenviament

After performing MAC learning, the server determines the forwarding action based on the destination MAC address of the encapsulated frame. The destination MAC address occupies octets 0–5 (inclusive, zero- indexed) of the Ethernet frame, i.e., octets 11–16 of the UDP payload.

Broadcast: If the destination MAC address equals `FF-FF-FF-FF-FF-FF`, the server MUST forward the PIXES packet (unmodified) to every client currently in the AUTHENTICATED state, excluding the originating client.

Known Unicast: If the destination MAC address is present in the MAC learning table and the associated session is in the AUTHENTICATED state, the server MUST forward the PIXES packet (unmodified) directly to that client's registered UDP address. This case is counted as a unicast forwarding event.

Unknown Unicast: If the destination MAC address is not present in the MAC learning table, the server MUST apply the configured Unknown Unicast Policy as described in Section 8.3.

8.3. Unknown Unicast Policy

Política per a Unicast Desconegut

The server MUST be configured with one of the following policies for handling unknown unicast frames:

`flood`: The server SHOULD forward the PIXES packet to all clients in the AUTHENTICATED state, excluding the originating client. This behaviour is equivalent to broadcast forwarding and allows the destination host to respond, enabling the server to learn the destination MAC from the reply.

`discard`: The server MUST silently drop the PIXES packet. The frame is not forwarded to any client. Traffic to as-yet-unknown destinations will fail until the destination MAC address has been learned via a previous inbound frame from the destination client.

In both cases, the server MUST increment a counter of unknown-unicast events for statistical purposes.

---

9. Security Considerations

Consideracions de Seguretat

PIXES provides no cryptographic protection. The following weaknesses are inherent to the protocol as specified and MUST be understood by operators before deployment:

1. **Cleartext transmission:** All Ethernet frames, including their payloads, are transmitted in cleartext inside UDP datagrams. Any on-path observer can read all traffic.
2. **Weak authentication:** The shared password is at most 8 ASCII characters. No salting, hashing, or challenge-response mechanism is used. The password is transmitted in cleartext in the AUTH message.

3. **No integrity protection:** An on-path attacker can modify PIXES packets, inject frames, or replay captured TRAFFIC messages. The sequence number field is advisory and is not cryptographically authenticated.
4. **No anti-spoofing:** The server associates a CID with a UDP source address, but this association can be overwritten by a REGISTER message from any source. A malicious host can hijack a session by sending a REGISTER with a victim's CID.

PIXES SHOULD NOT be used on untrusted networks without an additional security layer. Operators SHOULD consider encapsulating PIXES traffic within an encrypted tunnel (e.g., WireGuard [WireGuard] or IPsec [RFC4301]) when deploying over the public Internet or any network where confidentiality or integrity is required.

---

10. IANA Considerations

Consideracions IANA

This document has no IANA actions. PIXES does not use a registered UDP port number; the port is configured locally by the operator. PIXES is an Experimental protocol and does not request any IANA registry entries.

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11. References

Referències

11.1. Normative References

Referències Normatives

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11.2. Informative References

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Appendix A. Authors' Addresses

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